

# AI Output Critique (Part 6: Task A)

Critical Evaluation of LLM-Generated Test Strategy & Test Suite

**Assessment Part:** Part 6 Task A (AI Critique) **Target Document:** Generated Test Plan & Top 20 Cases **Date:** August 31, 2026

## Executive Overview & Verdict

This critique evaluates the output generated by an LLM when fed the Part 1 Wallet Transfer Limits specification. While the LLM demonstrated outstanding analytical capability in uncovering hidden architectural ambiguities (scoring high in risk analysis), it suffered from hallucinated test steps, invalid limit calculations, priority inflation, and structural ordering misalignments. Human engineering intervention remains essential to fix these critical flaws before implementation.

## 1. LLM Performance Scorecard

| Generated Dimension                | Rating             | Key Takeaway                                                                                                                            |
|------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Ambiguity Discovery (Questions)    | KEPT (9.5/10)      | Exceptional risk analysis. LLM independently spotted timezone exploits, pending race conditions, and limit inflation via reversals.     |
| Matrix Reduction Strategy          | KEPT (9.0/10)      | Smart deduplication logic. Collapsed 60 combinations to ~20 by isolating exempt paths and treating transfer types orthogonally.         |
| Test Case Precision & Calculations | CORRECTED (6.5/10) | Included invalid transfer amounts (TC-01/TC-04), vague assertion strings, and unrealistic manual steps (waiting for midnight in TC-11). |
| Structural & Wave Alignment        | REJECTED (6.0/10)  | Violated its own stated preamble wave sequence and suffered from heavy priority inflation (10 out of 20 marked P1).                     |

## 2. Deep-Dive: What to Keep vs. What Was Hallucinated / Flawed

### 1. Generated Elements Kept (High Value Added by LLM)

- Q1–Q10 Open Questions:** The model successfully inferred non-obvious business risks from silent specification rules (e.g., Q1 timezone manipulation, Q3 database locking race conditions, Q4 cross-day limit inflation via reversals).
- Risk Wave Prioritization (Test Plan):** The concept of grouping tests into Waves based on financial loss risk (Wave 1 Core Limits → Wave 2 Exemption/Reset → Wave 3 State Transitions → Wave 4 Scheduling/Concurrency) provided a solid strategic framework.
- Matrix Reduction Logic:** Correctly deduced that exempt paths ("Own Linked Bank Account") do not require cross-testing across all tiers and states.

## 2. Generated Elements Corrected or Rejected (Flaws & Hallucinations)

- **Incorrect Transfer Math & Step Logic (TC-01, TC-04):** In TC-01, the LLM had the user transfer 15,000 to leave 35,000 remaining, but failed to clearly separate testing the 20,000 per-transaction cap vs. the 50,000 daily cap. In TC-04, it grouped three transfers into one step without explicitly asserting incremental daily limit reduction.
- **Unrealistic Execution Logic (TC-11):** The LLM generated a test step asking a tester to "wait across midnight from 23:59 to 00:01". This was modified to require clock mocking/API time-override.
- **Vague Assertions (TC-03, TC-05, TC-20):** The LLM repeatedly output "rejected with a clear error." This was corrected to assert explicit HTTP status codes (HTTP 422) and concrete JSON payloads (`{"error": "LIMIT_EXCEEDED", "remaining": 40000}`).
- **Wave Sequence Contradiction:** While the LLM stated in its preamble that tests follow Wave 1 → 2 → 3 → 4, it output TC-17 & TC-18 (Wave 3) after TC-15 & TC-16 (Wave 4), and placed TC-20 (Wave 1 error validation) at the very end as a P2.
- **Priority Inflation:** The LLM tagged 10/20 cases as P1 and 8/20 as P2. This was rebalanced to a realistic distribution (~25% P1, 50% P2, 25% P3).

## 3. Where AI Helps vs. Harms Test Design

### Where AI Accelerates QA (The Benefits):

- **Rapid Edge Case Ideation:** LLMs excel at probing unstated assumptions, rapidly identifying gaps, and generating thorough open questions for product teams.
- **Drafting Boilerplate Test Suites:** LLMs generate large volumes of structured markdown/tables in seconds, drastically reducing initial typing effort.
- **Combinatorial Strategy:** LLMs are highly effective at explaining matrix reduction techniques and structuring risk-based test waves.

### Where AI Fails QA (The Dangers):

- **Boundary Math & Exact Values:** LLMs struggle with multi-step arithmetic, often miscalculating remaining limits or mixing up boundary inputs.
- **Real-World Execution Context:** LLMs lack operational context, generating impractical test steps like asking human testers to wait real time until midnight.
- **Contextual Ordering & Priority Rules:** Models frequently lose track of structural rules defined in their own preambles, leading to misaligned sequences and inflated priorities.

## 4. Prompt Engineering Lessons & Human Validation Workflow

**Validation Mandate:** Never deploy AI-generated test suites directly to execution without human oversight. QA Engineers must perform double-pass verification on arithmetic boundaries, assert exact error codes over generic prose, and enforce strict priority distribution.